CARVEDILOL

Take with food

HFrEF mortality benefit

CHEMICAL NAME (RS)-1-(9H-carbazol-4-yloxy)-3-{[2-(2-methoxyphenoxy)ethyl]amino}propan-2-ol

CLASS Non-selective β-blocker with α1-blocking activity

TRADE NAMES Coreg · Dilatrend · Eucardic · Carloc

THERAPEUTIC USES / INDICATIONS

- Heart failure with reduced EF (↓ mortality).
- Hypertension.
- LV dysfunction after MI.
- Portal hypertension (variceal prophylaxis).

ADMINISTRATION

- **PO** twice daily with food (reduces orthostatic hypotension).
- Titrate slowly every 2 weeks.

DOSAGES

- **Heart failure:** 3.125 mg twice daily; double every 2 weeks to target 25 mg twice daily (50 mg twice daily if >85 kg).
- **Hypertension:** 6.25–25 mg twice daily.
- **Portal hypertension:** 6.25–12.5 mg daily.

PHARMACOKINETICS

- **Onset:** 1 h. **Peak:** 1–2 h. **Duration:** 12 h.
- Bioavailability ~25% (first pass); metabolised by CYP2D6, 2C9.
- Excreted in bile/faeces. **t½:** 7–10 h.

PHARMACODYNAMICS

- **β1 + β2 blockade** → ↓ HR and contractility.
- **α1 blockade** → vasodilation, ↓ afterload.
- Antioxidant properties.

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Dizziness, postural hypotension.
- Fatigue, bradycardia.
- Weight gain, oedema.

ADVERSE EFFECTS (SERIOUS)

- Worsening HF during titration.
- Bronchospasm.
- AV block.
- Masked hypoglycaemia, hyperglycaemia.

CONTRAINDICATIONS & CAUTIONS

- Decompensated HF needing IV inotropes.
- Asthma or bronchospastic disease.
- Severe bradycardia, 2nd/3rd-degree heart block, sick sinus, cardiogenic shock.
- Severe hepatic impairment.

KEY INTERACTIONS

- **Verapamil, diltiazem, digoxin, amiodarone:** bradycardia.
- **Rifampicin:** ↓ levels.
- **Ciclosporin:** ↑ ciclosporin.
- **Insulin:** masked hypoglycaemia.

NURSING MANAGEMENT

- Check HR and lying/standing BP before each dose.
- Give with food.
- HF: daily weights, fluid status.
- Hold if HR <55 or symptomatic hypotension (per order).
- Taper on stopping.

PATIENT EDUCATION

- Take with meals.
- Stand up slowly; sit if dizzy.
- Weigh daily; report weight gain or breathlessness.
- Do not stop suddenly.

KEY POINTS

- β + α1 blocker; vasodilating.
- Postural hypotension; give with food.
- HFrEF target 25 mg BD.
- Avoid in asthma.